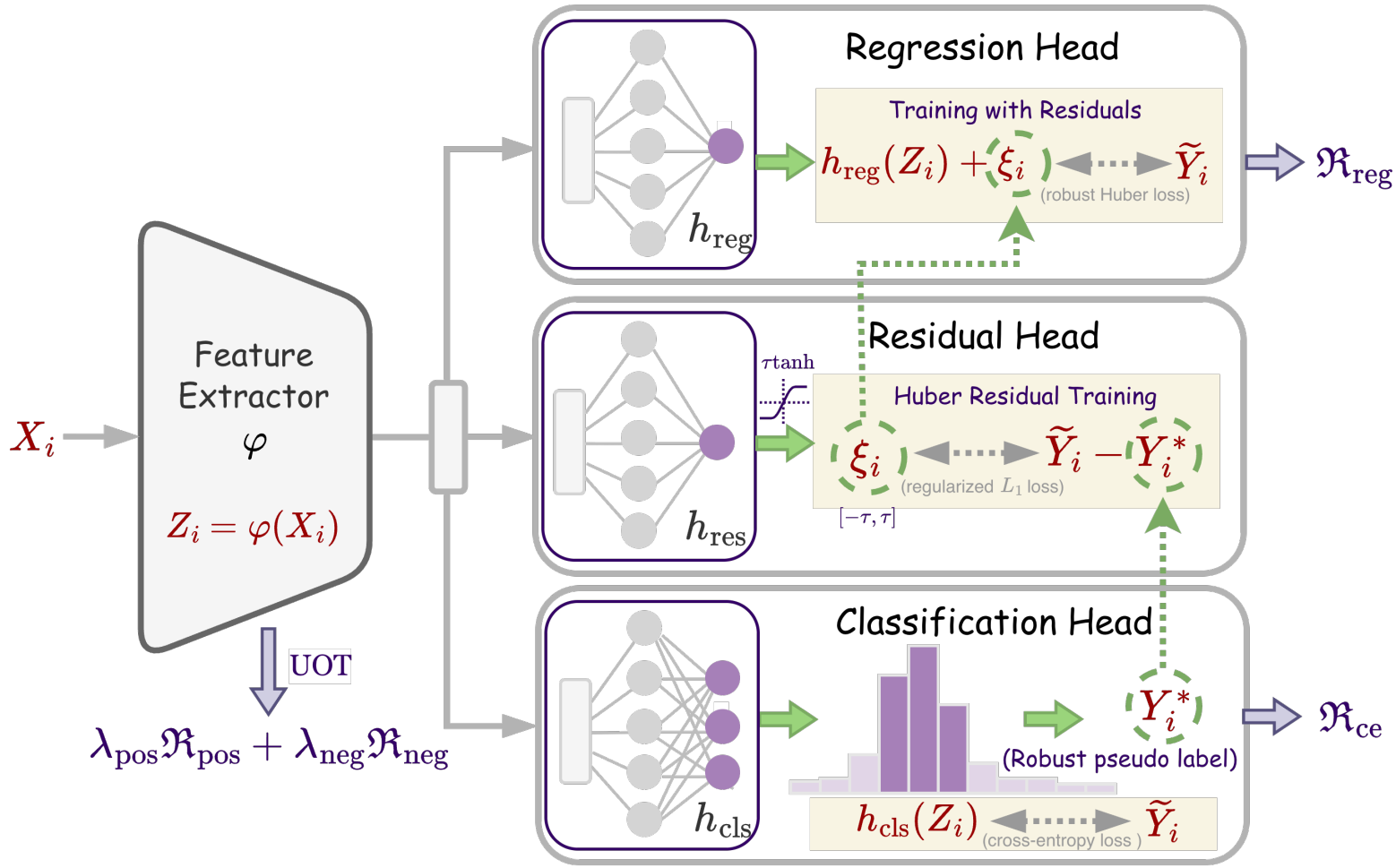
Regression Head
Training with Residuals
$h_{\text{reg}}(Z_i) + \xi_i$
$\widetilde{Y}_i$
(robust Huber loss)
h_{reg}
$\mathfrak{R}_{\text{reg}}$
Residual Head
$\tau\tanh$
Huber Residual Training
ξ_i
$\widetilde{Y}_i - Y_i^*$
(regularized L_1 loss)
$[-\tau, \tau]$
h_{res}
Classification Head
Y_i^*
(Robust pseudo label)
$h_{\text{cls}}(Z_i)$
(cross-entropy loss)
$\widetilde{Y}_i$
h_{cls}
$\mathfrak{R}_{\text{ce}}$
X_i
Feature Extractor
φ
$Z_i = \varphi(X_i)$
UOT
$\lambda_{\text{pos}}\mathfrak{R}_{\text{pos}} + \lambda_{\text{neg}}\mathfrak{R}_{\text{neg}}$